

NAM Service Accelerator(LPM) new features

April, 2021



3DEXPERIENCE®

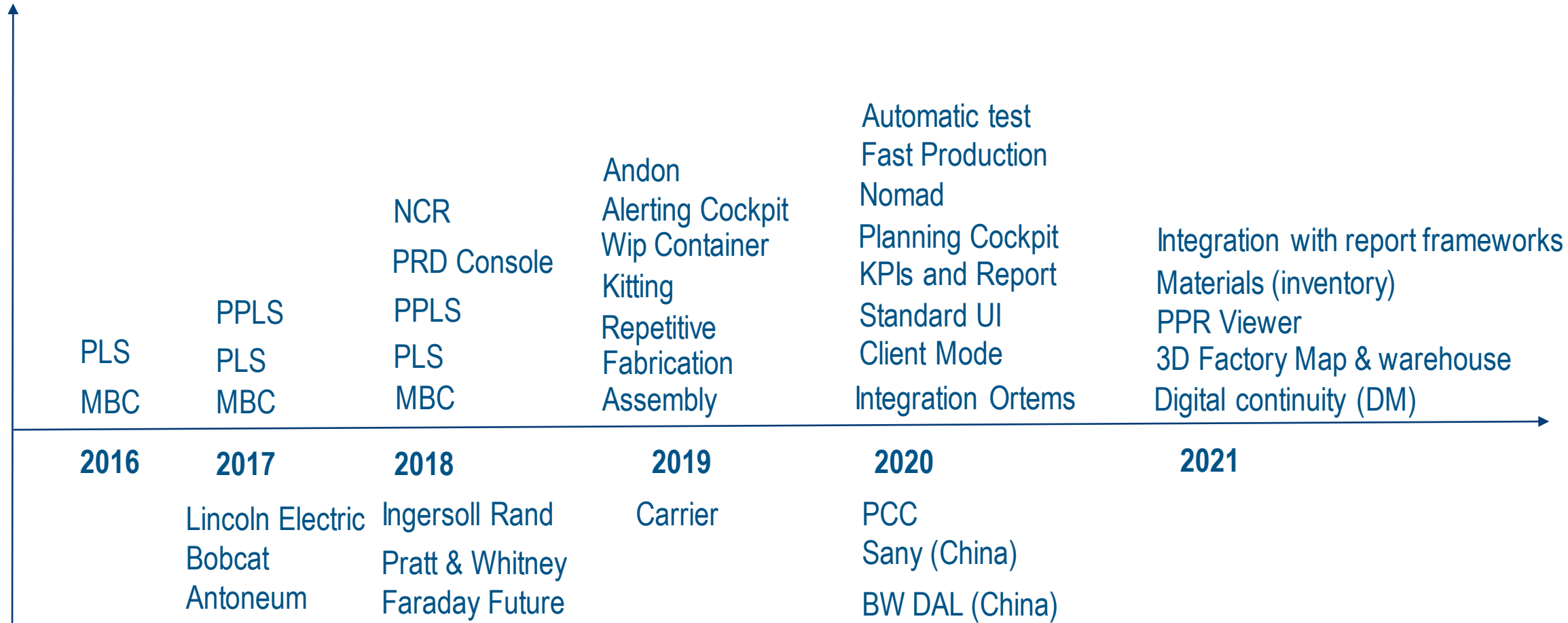
Agenda

1. LPM Landscape
2. Supported Use Cases
3. Additional New features

What is LPM

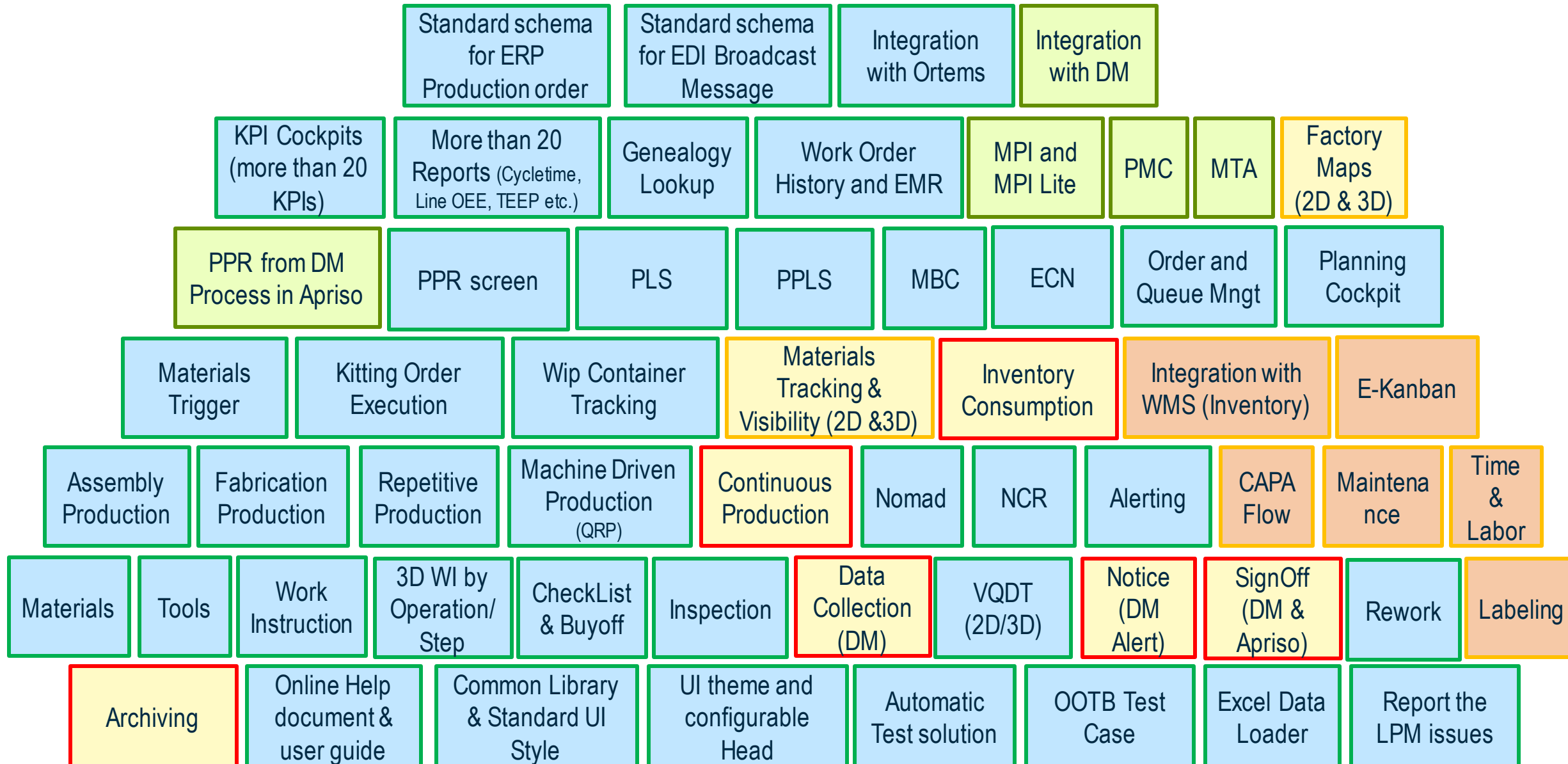
- ▶ LPM is a NAM service asset which supplies the prepackaged solutions to
 - ▷ Supply the start kit of the project and speed up the project implementation
 - ▷ Supply demo kit for the presales
 - ▷ Supply the visualized solution during work shop phase etc.
- ▶ We collect the LPM's solution requirements from actual project or RFP
- ▶ We reuse the good features from the actual projects
- ▶ We utilize the resource on the bench and train the team as well

LPM Evolution

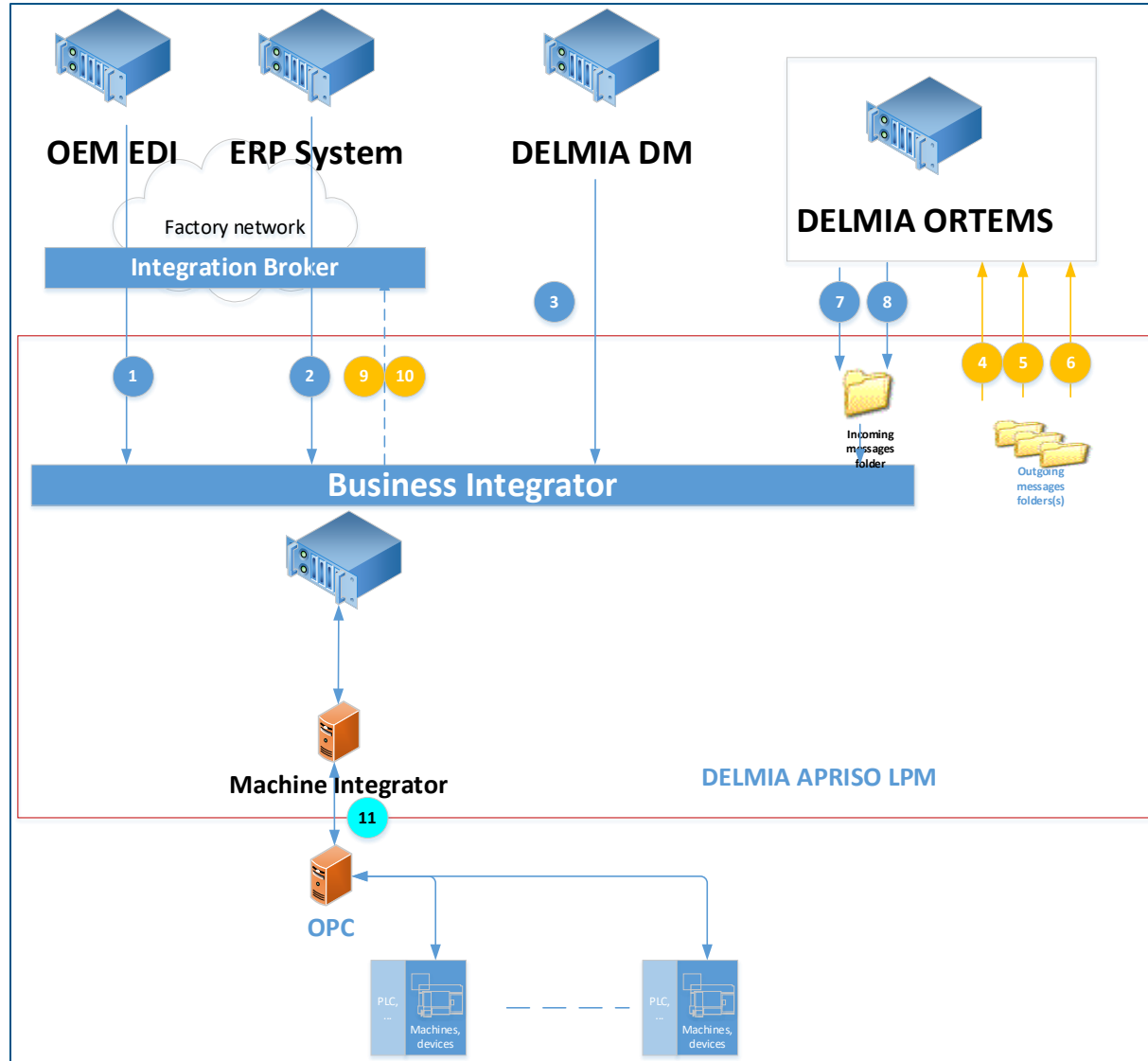


High Level LPM functional Landscape

LPM OOB
Platform/Apriso OOB
LPM Partial / in process
LPM future

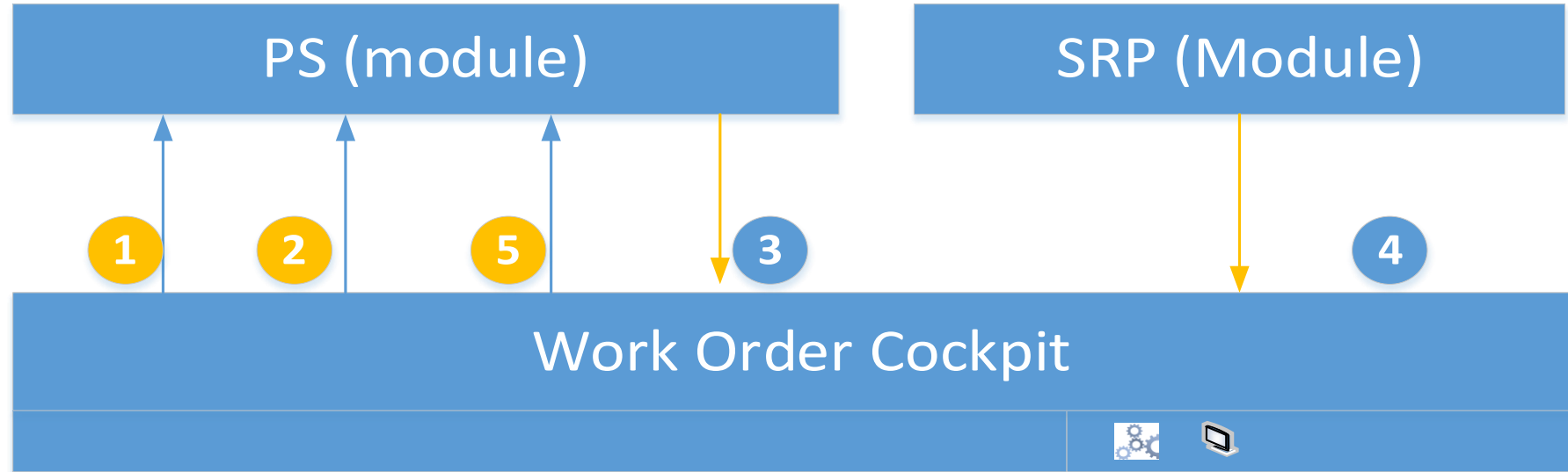


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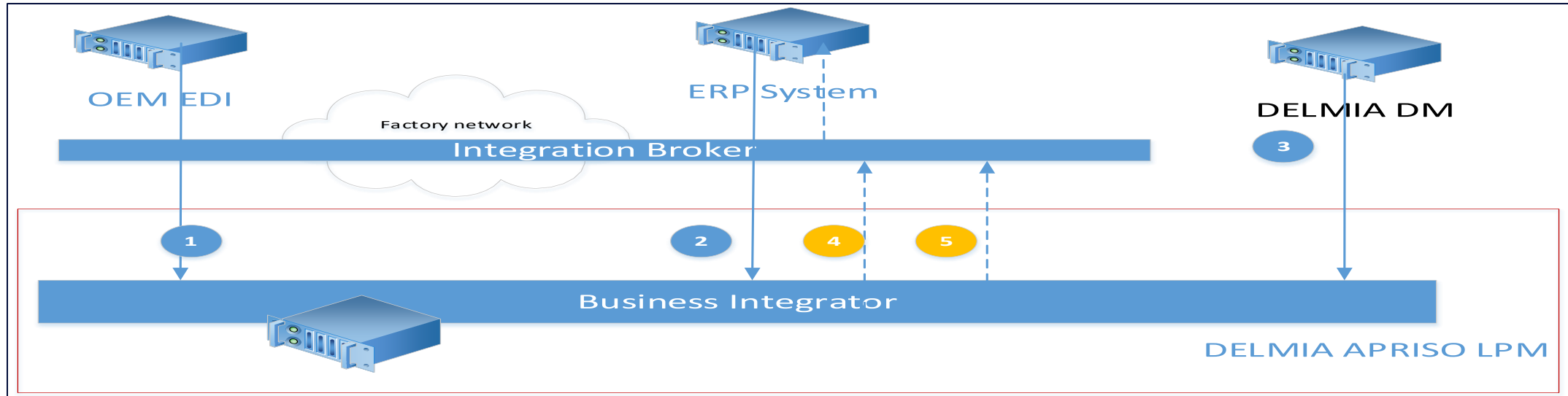
1. Production Requirement (BroadCast Message from OEM) - Schema
2. Production Order (Routing, Components, WI, etc.) – Schema
3. Process from DM (OOB)
4. Production orders and Operations (Apriso -> Ortems)
5. Maintenance Order (Apriso -> Ortems)
6. Operation status (Apriso -> Ortems)
7. Scheduled Production order / Maintenance Order and Operations (Ortems - > Apriso)
8. SRP generated Production order with Routing and components (Ortems -> Apriso)
9. Operation Completion Message (Apriso - > ERP)
10. Order Completion Message (Apriso- > ERP)
11. Machine Integration via OPC

Integration with Ortems



Number	Type	Description	Direction	Transaction	Message format
1	Outbound Message	Uncompleted Production orders and Operations	Apriso - > Ortems	Background Jobs	Csv
2	Outbound Message	Uncompleted Maintenance Order	Apriso - > Ortems	Background Jobs	csv
3	Inbound Message	Planned Work order and Operations	Ortems -> Apriso	BI (Incoming Message)	csv -> xml
4	Inbound Message	SRP generated Work order with Routing and components	Ortems -> Apriso	BI (Incoming Message)	csv -> xml
5	Outbound Message	Operaton status	Apriso -> Ortems	Background Jobs	csv

Standard schema for ERP /EDI



	Type	Description	Direction	Transaction	Message format
1	Inbound Message	The BroadCast Message interface is used to receive the OEM broadcasted message for materials requirement.	OEM EDI -> Apriso	BI (Incoming Message, Webservice call)	XML
2	Production order Interface	The Production Order interface is used to download the Production Order, and the routing information from the ERP system to Apriso	ERP -> Apriso	BI (Incoming Message, Webservice call)	XML
3	Process Interface	The Process from DELMIA DM	DM -> Apriso	BI (xPDM)	XML
4	Outbound Message	Operation Completion Status	Apriso - ERP	BC generated xml / BI	XML
5	Outbound Message	Work Order Completion Status	Apriso - > ERP	BC generated xml / BI	XML

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1. LPM Landscape

2. Supported Use Cases

3. Additional New features

LPM use cases

- ▶ Assembly Production
- ▶ Fabrication Production
- ▶ Repetitive Production
- ▶ Machine Driven Fast Production
- ▶ Nomad Production Use Case

Assembly Production Use Case

-  New/Updated Feature
-  Existed Feature



Updated Production Console

Variable production console layouts by configuration

Order information, stream work center, takt time, and machine state

Small icons or big icons of activity, Activity sequence can be configured

Activities show in one view or separated view

The image displays two screenshots of the DELMIA Apriso Production Console interface, illustrating its variable layouts and configuration options.

Left Screenshot: Shows the main console layout. The top bar displays order information (8592 Order, PC-CASE Product, 0010 Operation, S7048 / / Serial/Lot/Container, 1 - Step Description, 0 / 2 Qty, 12/11/2021 4:00:00 PM Due Date). The left sidebar contains icons for Team, Order List, Performance, Alert, Work Instruction, Materials, Tools, Non Conformance, Notes, and Containers. The main area shows a 3D model of a hand brake lever component.

Right Screenshot: Shows a detailed view of a work center. The top bar displays order information (6175 Order, 120-001 Product, 0010 Operation, S2537 / Serial/Lot, 1 - Step Description, 9 / 10 Qty, 6/3/2019 2:00:00 PM Due Date). The main area is divided into sections: Resources (Tools Description, Qty Available, Used Qty, TOOLS), Work Instruction (showing a 3D model of a car chassis), and Quality (showing a table with columns: Characteristic..., UOM, COLLECTVALUE, Last Value, Comment). The Quality table shows a row for Torque (0 / 5) with a value of 15.000 and a comment of 15.000. The bottom bar includes buttons for Start, Pause, Complete, and a Scan field.

3D model shows by Step and components

The screenshots illustrate the 3D model's ability to show components by step. The first screenshot shows the '1 - (Load Cushion Pan)' step, the second shows '2 - (Load Frame)', and the third shows '3 - (Load cushion Frame)'. Each step highlights a different component of the assembly.

Product Description Table:

UOM	SerialNo/LotNo	Quantity	Total/ Kitted/ Remaining	Reference
GCushion Pan_Exercise-34		1	0000/0001/000	

with 3D model, show only the part that defined for the selected step

Rework Flow – Serial Tracked

Rework Flow for Serial Tracked Production

NCR Analysis
Disposition: Rework



Create the Rework Order



Edit the Rework Order



Complete the activates and
complete the operation

NCR will be closed if rework order
is completed



Components to removed



Rework Task

Genealogy Lookup and Work Order Activity

LPM DELMIA Apriso Genealogy Lookup 10:30:31 AM Rocky Wang

Product #: 121-001_1.0 Serial #: Lot No: Order Number:

PRODUCT NO	SERIAL NO	LOT NO
121-001	S209	
120-001	SN00001	
Parent		
100-001		LT0001
Parent		
100-002		LT0002
Parent		
100-003		LT0001
Parent		
102-108		
Parent		
105-001		
Parent		

GENERAL WORK IN PROCESS HISTORY



Product 121-001
Product Description Body
Product Revision 1.0
Inventory Type Stock material
Serial # S209

Genealogy Detail

	Relation	Serial & Lot #	Product	Wip Order #	Order type	Operation #	Step #	Quantity
☐	Child	SN00001	120-001 - 1.0	1063	1	0010	2	1.000
☐	Child	LT0001	100-001 - 1.0	1063	1	0010	2	1.000
☐	Child	LT0002	100-002 - 1.0					
☐	Child	LT0001	100-003 - 1.0					
☐	Child		102-108 - 1.0					
☐	Child		102-108 - 1.0					
☐	Child		102-108 - 1.0					
☐	Child		102-108 - 1.0					

LPM DELMIA Apriso Genealogy Lookup 4:21:52 PM Rocky Wang

Product #: 121-001_1.0 Serial #: Lot No: Order Number:

GENERAL WORK IN PROCESS HISTORY

Profile: Default Profile

Activity	Activity Item01	Activity Item02	Activity Item03	Activity Item04
stopoperation	QRC_EQU0010	Not filtered	Not filtered	Not filtered
stopoperation	QRC_EQU0010	Not filtered	Not filtered	Not filtered
stopoperation	QRC_EQU0010	Not filtered	Not filtered	Not filtered
stopoperation	QRC_EQU0010	Not filtered	Not filtered	Not filtered
startoperation	QRC_EQU0010	57265		
order_explode	PPL5	161	Work Order Cockpit	QRC_M
order_create	QRC_MainLine	100000178	10	

Wip Order No: 9401 Order Type: Production Order

Product No: 121-001 Description: Body

Operation Sequence: Step Sequence No: 0 Work Center:

Parent Wip Order No: Parent WipOrderType:

Parent Operation Seq No:

Facility: C1P1 Production Line: QRC_MainLine

ProductNo: 121-001 Order Description:

Priority: 1 Lot #:

Original Status: Work Order Status: New

Order Quantity: 10 UOMCode: EA

Process Plan: 121-001(QRC_MainLine)

Due Date: 3/31/2021 12:00:00 AM

Scheduled Start Date: 1/1/1900 12:00:00 AM Scheduled CompletionDate: 1/1/1900 12:00:00 AM

Expected StartDate: 1/1/1900 12:00:00 AM Expected CompletionDate: 1/1/1900 12:00:00 AM

Selected records: 1

LPM use cases

- ▶ Assembly Production
- ▶ Fabrication Production
- ▶ Repetitive Production
- ▶ Machine Driven Fast Production
- ▶ Nomad Production Use Case

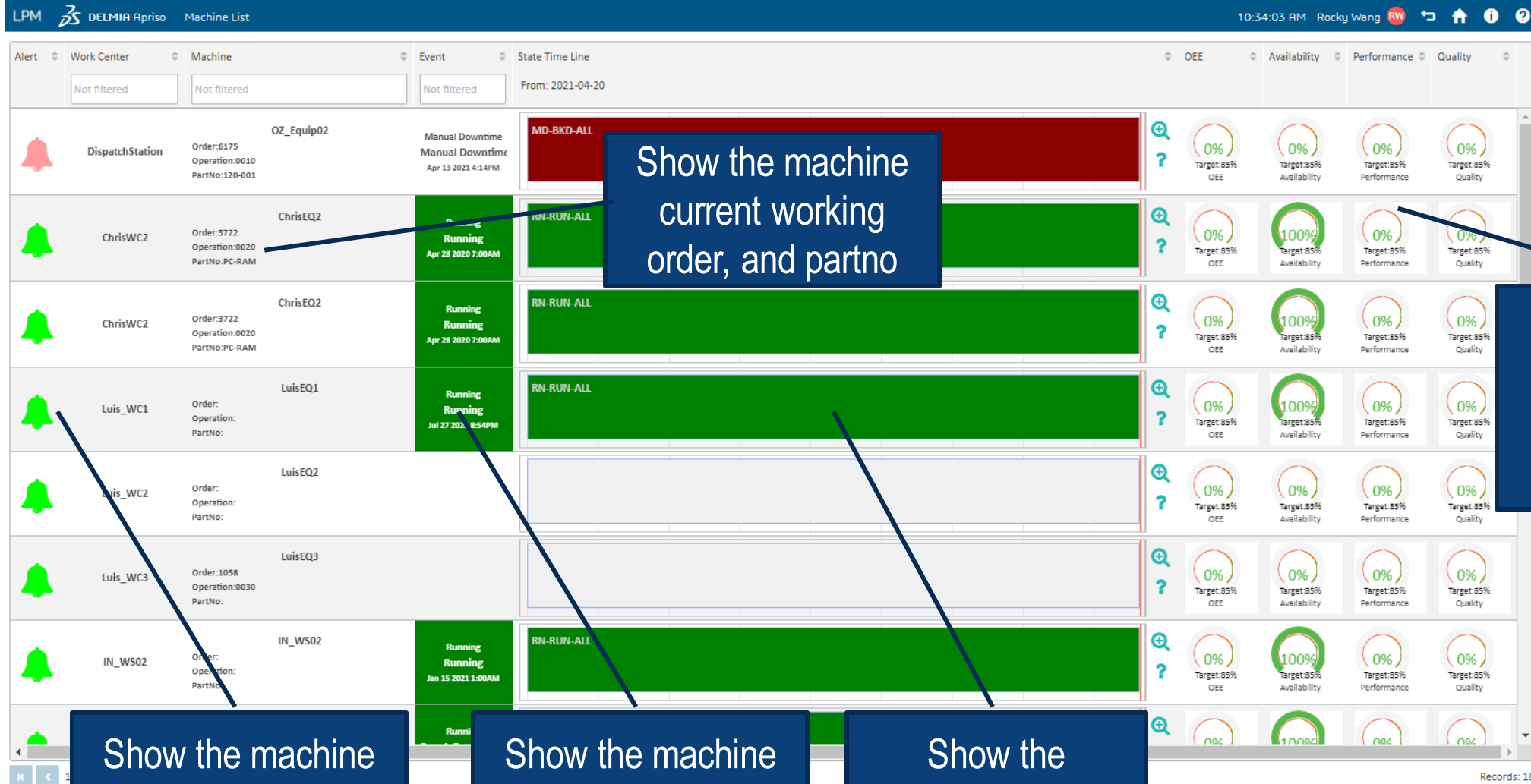
Fabrication Use Case

X New/Updated Feature

X Existed Feature



Updated Machine List



Show the machine current working order, and partno

Show the Machine OEEs KPI, update by interval (5 min)

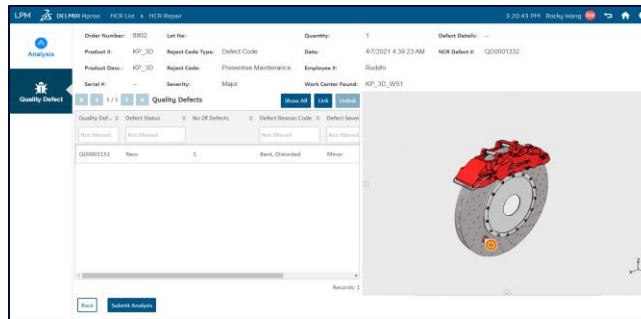
Show the machine current alert status – update automatically by AMB

Show the machine current state – update automatically by AMB

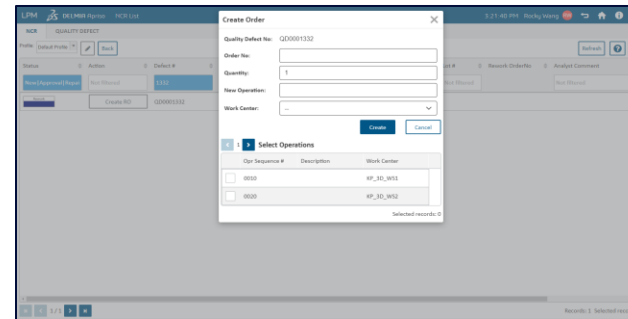
Show the Machine State time line of the shift or day

Rework Flow – Non Serial Tracked

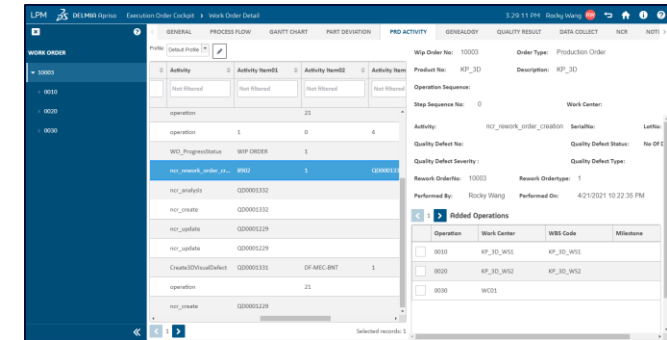
Production order Split and rework flow of Non Serial tracked production



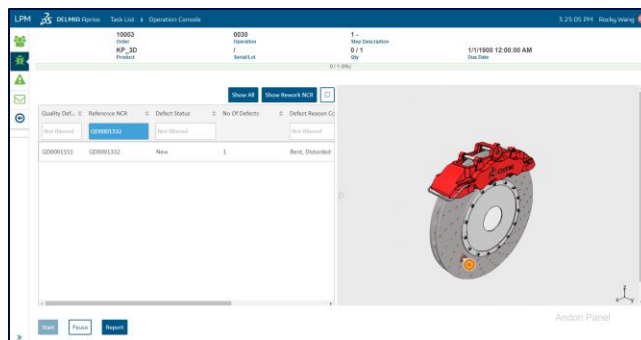
NCR Analysis
Disposition: Rework



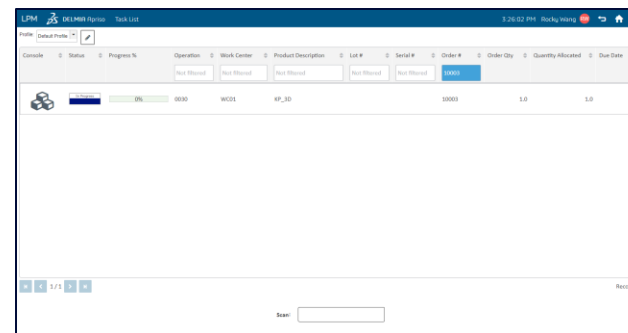
Create rework order, and
split original order



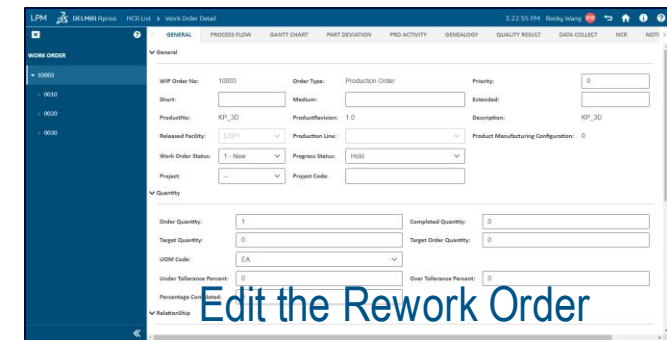
Copy activity history and
EMR from original order
to new order



Execute the Rework Order
Close NCR when rework order is closed



Start the Rework Order from Task List

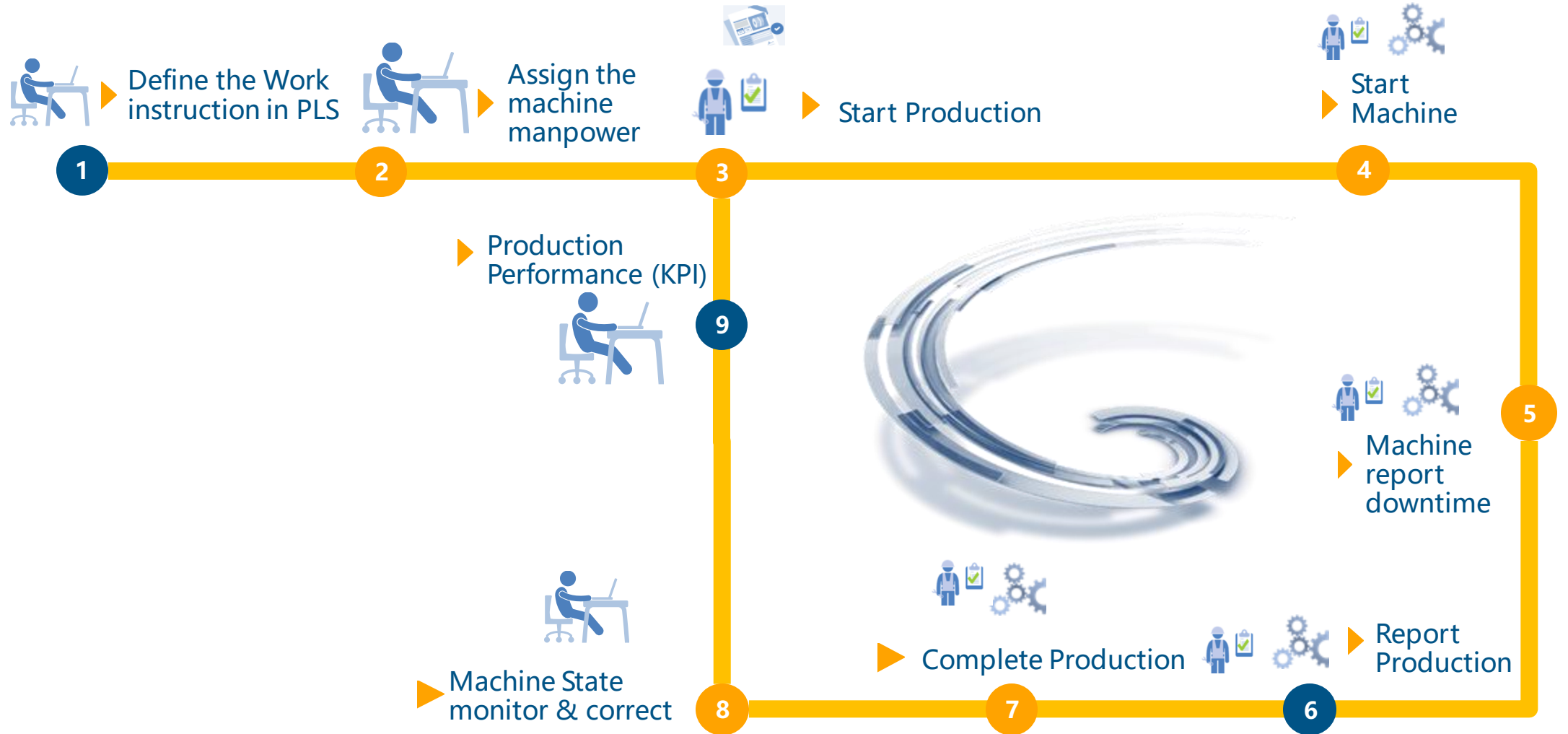


Edit the Rework Order

LPM use cases

- ▶ Assembly Production
- ▶ Fabrication Production
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- ▶ Nomad Production Use Case

Repetitive Use Case



Updated Repetitive Console

Show the working parts and progress

Show the machine KPI



Machine state and change state

Show the work instruction

Start/Stop/Report the production
Supply new Lot during report

LPM use cases

- ▶ Assembly Production
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Machine Driven Fast Production

Quick response production

- ▶ The Production has the short cycle time (less than 40s by cycle time)
- ▶ Most production console is updated by the machine signal
 - ▷ Cycle Start – Update order, download run book etc.
 - ▷ Step Complete – Update the work instruction
 - ▷ CycleEnd – Materials consumption, update quantity, progress
 - ▷ Inspection Data Collection
 - ▷ Serial Component consumption
- ▷ Machine State Change
- ▶ Additional features:
 - ▷ Introduce the Unit at first station, or dispatch the unit from production Queue
 - ▷ Load Materials to machine
 - ▷ Pack the unit at last work station
 - ▷ Report NCR
 - ▷ Leave Notes (Alert)
 - ▷ Simulator for testing
 - ▷ KPIs and Reports for monitoring production

Machine Driven Fast Production Use Case



Machine Driven Fast Machine Console

LPM  DELMIA Apriso Machine List > Quick Response Console 3:04:20 PM RW

1415 Order
PC-CASE Product

0010 Operation
Serial #

Assemble the power unit to the PC ...
Operation Description
10 / 10 Qty

171 OWS01 Station Buffer

45 OWS02 Next Station

100% Target 85% OEE

100% Target 85% Availability

100% Target 85% Performance

100% Target 85% Quality

RUNNING Since: 2021-04-15 00:00

Order Information

Previous, Current and Next Station KPI

Station OEEs and State

Steps

Step work instruction

Materials

Product #	Qty	
PC-Frame	1.0...	LT00010-001
PC-PU-FAN	2.0...	LT00010-001
PC-PU-Convert	1.0...	LT00010-001

Collected Process Data

Characteristic	Inspection Va...
Materials and Quality data	

Andon Panel

Nonconformance Notices



Pack Station

Station KPIs

16

LPM  DELMIA Apriso Machine List Quick Response Console 4:00:59 PM Rocky WANG RW

1415 Order
PC-CASE Product

0100 Operation
S313 Serial #

Operation Description
8 / 10 Qty

OWS09 Previous Station Production 70%

46 OWS10 Station Buffer

100% Target: 85% OEE

100% Target: 85% Availability

100% Target: 85% Performance

100% Target: 85% Quality

RUNNING
Since: 2021-04-15 00:00

Product #	Serial #	Order Number
Not filtered	Not filtered	Not filtered
PC-TOWER	S2	2
PC-CASE	S25	214
PC-CASE	S27	215
PC-CASE	S30	216
PC-CASE	S314	1415

< 1 > Selected records: 0

Container L5 2

Scan Unit SerialNo S314

Nonconformance Notices New Container Complete Container

Materials

Product #	Qty	Identity #
-----------	-----	------------

Packed Container and Unit

Previous working unit and Current Station buffer

Complete the Container

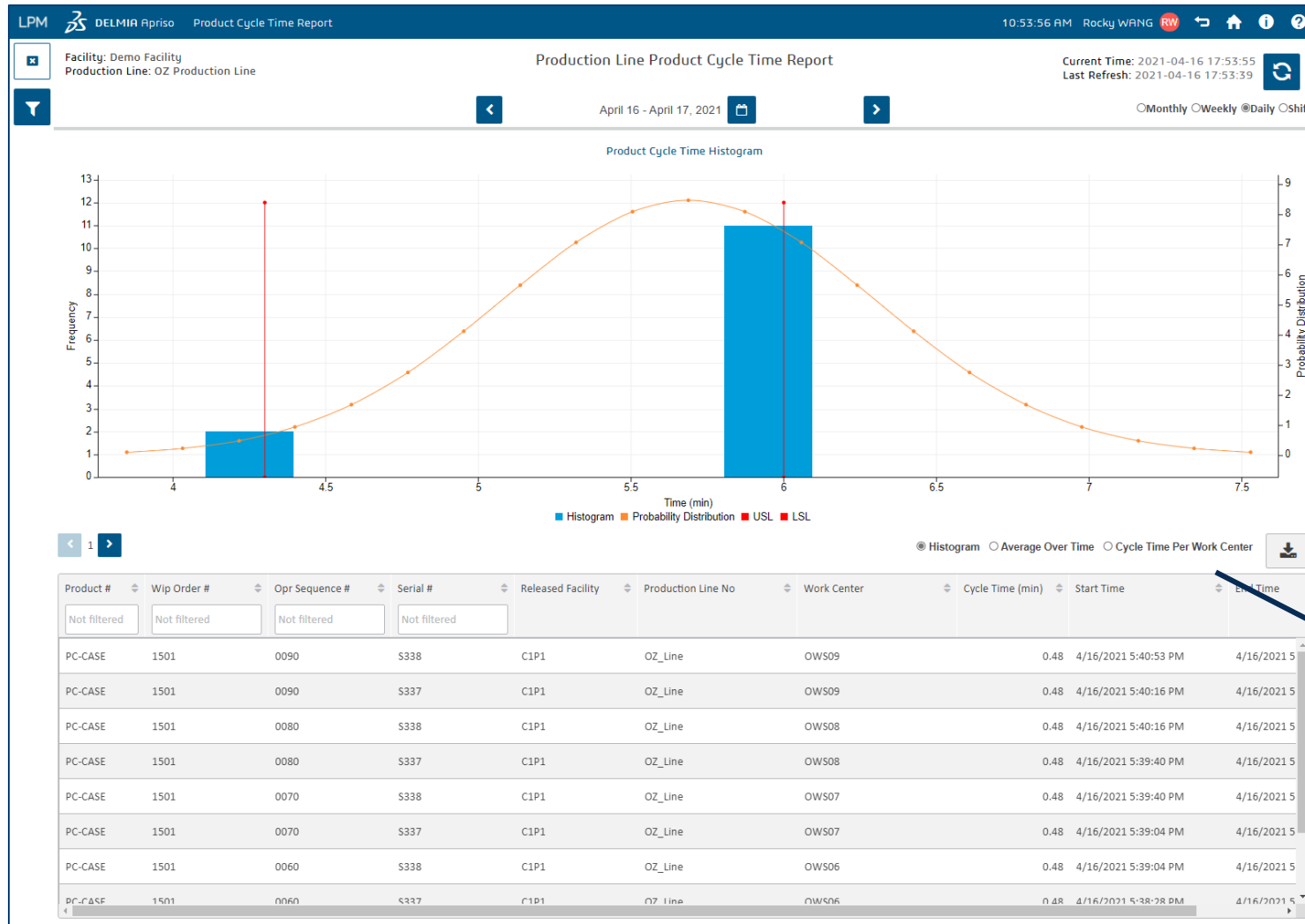
Andon Panel

Machine State

Previous working unit and Current Station buffer

Complete the Container

Production Cycle Time Analysis

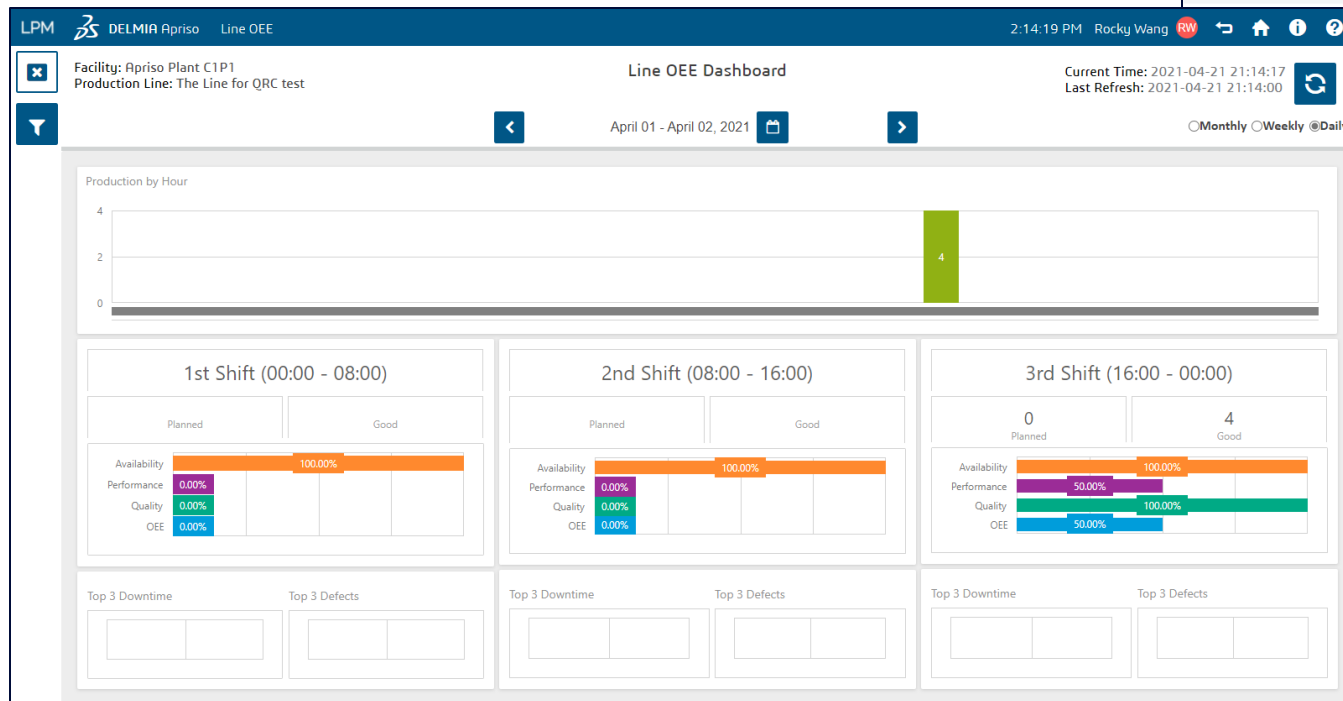


- Production Cycle time Histogram analysis
- Production Cycle time by the operation, line

The different ways to display detail data

Line OEE and Genealogy

- Line OEE to show the OEE of the Line for shift



The screenshot shows the 'Genealogy Lookup' interface. It displays a hierarchy of product, serial, and lot numbers. The selected item is '121-001-19020704'. The right panel shows details for this item, including 'Wip Order No: 109', 'Order Type: Production Order', 'Product No: 120-001', 'Description: RTR5 Test Product with Visualization', 'Operation Sequence: 0070', 'Step Sequence No: 1', 'Work Center: FA_WS02', 'Serial No: S60', 'Lot No: 21201', 'Sampling: TT_ANGLEVALUE(1.0)', 'Description: Angle', 'USL: 150.00', 'LSL: 20.00', 'UCL: 135.00', 'Collected Value: 34', 'Collected Attribute: 34', 'Performed By: Rocky Wang', and 'Performed On: 2/8/2019 1 AM'. The bottom panel shows a list of activities for the selected item, including 'operationstep', 'operation', 'component', and 'quality'.

Product NO	SERIAL NO	LOT NO
Parent		
120-001	121-001-19020703	
121-001	121-001-19020704	
Parent		
120-001	121-001-19020704	
121-001	121001-3602-01	
121-001	210010211	

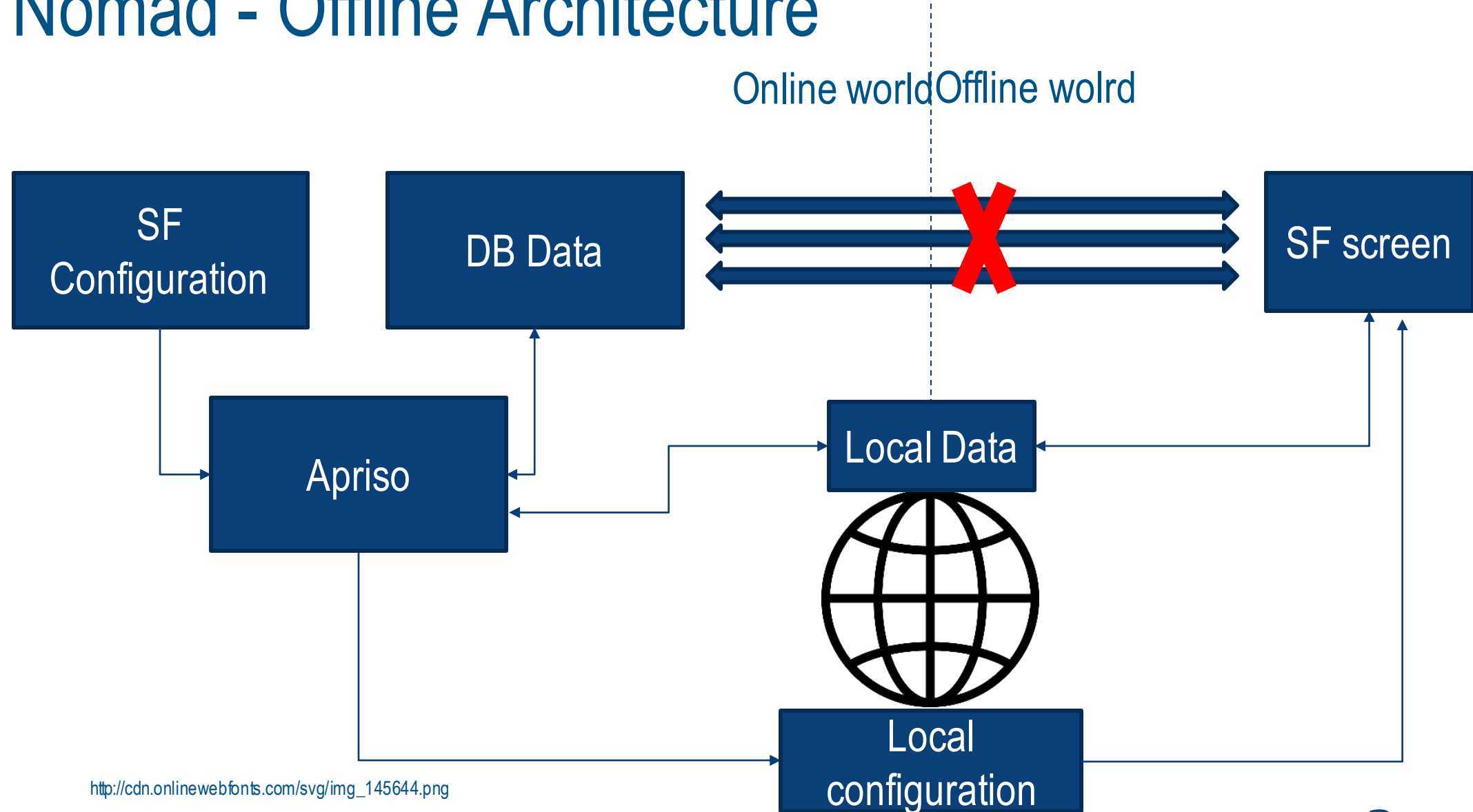
Activity	Activity Item01	Activity Item02
Not filtered	Not filtered	Not filtered
component	130-001	
operationstep	0	17
operation		21
operation	21	17
component	130-000	
operationstep	21	17
quality	TT_ANGLEVALUE(1.0)	34
quality	TT_TORQUEVALUE(1.0)	34
operationstep	0	21
operation		21

- Genealogy shows the Genealogy and production activities

LPM use cases

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Nomad - Offline Architecture



Nomad Production Use Case



Assign the task to the Employee

LPM DELMIA Apriso Production Planning Cockpit 2:18:23 PM Rocky Wang RW

Production Line Production Plan: CarAssembly

Employee List:

- UK KIOSK User
- RQ Rafael Quijano
- CP Chintankumar PATEL
- ST Shinha Tsuchida
- RT Rendy Tejada
- Julio TREVINO
- SB Samuel Brion BASKERVILLE
- RW Rocky Wang
- JL Julio Lopez
- JC Jorge CUERVO
- RK Ruddhi Kadam
- MK Madiha
- RS Rahul SAPAR
- RP Roshane Andre PASCUAL
- LG Luis Galicia
- CT Chris
- SH Syed Haider
- KC CHUNG Kenny
- RW Rocky WANG
- YL Yuta LOVAN
- KK Kyle KIM
- JC Jesus CORTEZ
- HY Harvy
- PD Phuc Doan
- AG Akshat Gulati
- KS Komal SARIN
- KW Katherine WANG
- BS Bill Scheppe
- 1R Oscar
- IR Ivan
- RT Ryan TEER
- FK Farajana Khatib
- ML Mikalie LAI
- DL Dennis Logan
- KP Karan PATEL

Time Period: Minutes | HalfHour | **Hour** | Quarter_Day | Half_Day | Day | Week | Month | Year | Save

20 April 2021

21 April 2021

Operations:

- CarAssembly
- DispatchStation
- OZ_Equip02
- DisptachStation
- Trim1_WS01
- Trim1_WS01
- Trim1_WS02
- Trim1_WS02
- Test_WC

Drag the available employee from the list and assign to operation

Drag to change the planned start and end time

Workspace display the available timeslot

Nomad Operation Console

Work for the IOS, Andriod app and SSLs desktop

The screenshot displays the Nomad Operation Console interface. The top bar shows 'LPM', 'DELMIA Apriso', and 'Nomad Console'. The status bar indicates 'Offline Production Console' and 'Online (It is downloading data from server! (12/7/2020, 10:45:38 AM))'. A blue notification bar states 'Downloading content - do not go offline.'.

The main interface has four buttons: 'Download Data', 'Tasks', 'Upload Data', and 'Back'. Callouts explain these functions:

- Download the task and data from the server**: Points to the 'Download Data' button.
- Go Offline and start the task**: Points to the 'Tasks' button.
- Upload the offline transaction data to Server when online**: Points to the 'Upload Data' button.

An inset window shows the 'Nomad Tasks' screen. It displays a task titled 'The Car of Lighting' with details: Operation 0010, OrderNo 7930, Work Center DispatchStation, Production Line No CarAssembly, Lot #, Serial # S6517, Order Qty 1 / 1, and Due Date 2020-12-14T08:00:00. A progress bar shows 4% completion. Callouts for this window include:

- Select the task to start the task**: Points to the task title.
- Task details**: Points to the task information fields.
- Task activities and status**: Points to the status icons at the bottom.

The Activities of the Task

LPM  DELMIA Apriso

Nomad Console Nomad Tasks Nomad Production Console

10:59:40 AM Rocky Wang

7930 0010
Order Operation
120-001 S6517 /
Product Serial/Lot

0 / 1 2020-12-14T08:00:00
Qty Due Date


Step List 0/1


Check List 0/8


Material 0/7


Work Instruction 4


Tools 1/3


Quality 0/3


3DXML 1


Non Conformance 0/0


Notes 0/0

Available activates of the task and its status

Back Start Hold Step Pause Complete

Agenda

1. LPM Landscape
2. Use Cases
3. Additional New features
4. Roadmap

Other new features in A20.02

- ▶ Standardize the UI and Common Library
- ▶ Convert the solution to support the Client Mode
- ▶ New Reports / KPIs (more than 30 KPIs and 20 Reports)
- ▶ Automatic Test solution
- ▶ Integrated help, and user Guides and implementation Guides

Standardize UI & Common Library

LPM Common Library, SVC CSS User Guide

- ▶ UI style defined in css file instead of view operation
- ▶ Common css class across LPM screens
- ▶ Themes

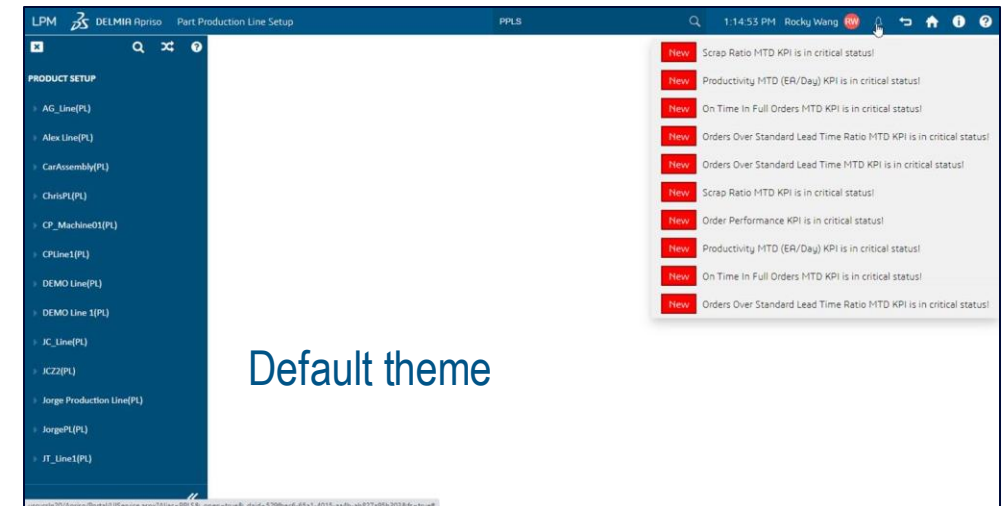
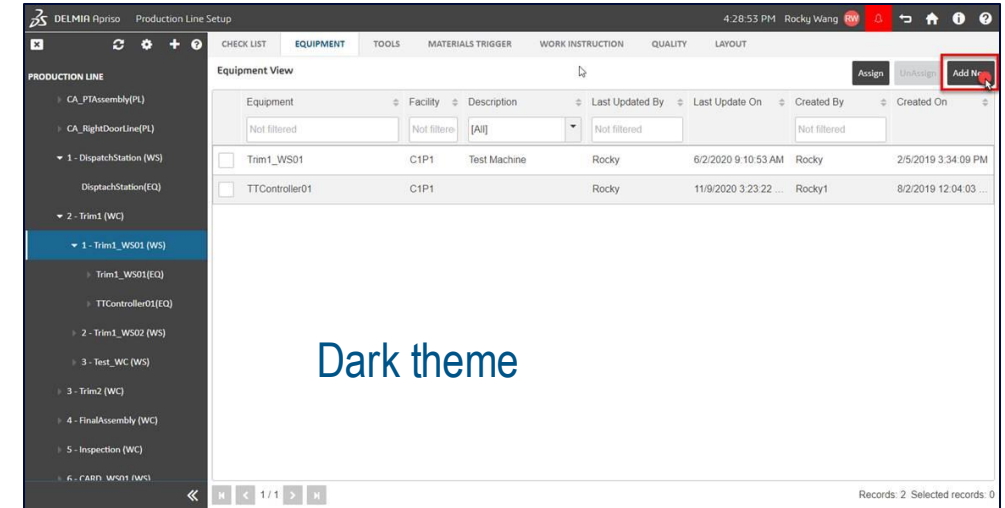
The screenshot displays the LPM SVC UI Documentation page. The top navigation bar includes the LPM logo, DELMIA Apriso, SVC UI Documentation, and a user profile for Rocky Wang. The left sidebar lists various UI components: Home, Typography, Icons, Popups, Buttons, Input, Textarea, Checkbox, Radio, Switch, Grid, Tree, and Utilities. The main content area is titled 'USAGE' and contains two sections: '1 Add CSS file' and '2 Use Component'. Below these is a 'STEP BY STEP GUIDE' section. The '1 Add CSS file' section explains that each component in the SVC UI Library is included in a CSS file, and the file link will be shown below the component documentation. The '2 Use Component' section explains that the CSS class, or HTML, for the component depends on where it can be used and where you would like to use it (view or operation). The 'STEP BY STEP GUIDE' section starts with 'Let's take a component and use it as an example to go through step by step.' and then '1 Select Component'. It explains that in this guide, the Button component will be used, and the user should open the Button component documentation (left sidebar) and follow the directions. A smaller version of the same documentation page is shown as an inset, with the 'Buttons' link in the sidebar highlighted with a red circle and the number '1'.

Configuration of the screen Themes and Head

SVC_SCREEN_CSS

SequenceNo	CSSAlias	Logo	OperationCode	Show Alert icon on	Hide the Search b	HideBack	HideHome	HideHelp	HideInformation
0	default			☐	☒	☐	☐	☒	☐
0	dark			☒	☐	☐	☐	☐	☐
0	dark	/Apriso/Portal1		☒	☐	☐	☐	☐	☐

Index	Field	Description
1	CSSAlias	select the Theme for the screen
2	Logo	 customized log which shows at the Left
3	OperationCode	can execute the sub operation which can do more that does not include in the configuration
4	ShowAlerticon	 Show or hide the alert icon on the head.
5	HideSearchBox	 Display or Hide SearchBox in the center of the head
6	HideBack	 Display or Hide Back button
7	HideHome	 Display or hide Home Button
8	HideHelp	 Display or Hide Help button
9	HideInformation	 Display or Hide about button



KPIs

► Equipment:

- ▷ Availability
- ▷ Performance
- ▷ Quality
- ▷ OEE
- ▷ TEEP
- ▷ Utilization
- ▷ Downtime
- ▷ Defects
- ▷ Downtime by the Reason Code
- ▷ Down Time by Day
- ▷ Downtime duration by Work Center, Reason Code

► Production:

- ▷ Target Quantity
- ▷ Actual Quantity
- ▷ Processed Quantity
- ▷ Rework Quantity
- ▷ Scrap Quantity
- ▷ Scrap Ratio
- ▷ Completion rate
- ▷ On Time
- ▷ FPY
- ▷ Time In Area
- ▷ Do what's Due

▷ Do what behind

- ▷ WIP
- ▷ Cycle Time
- ▷ Planned order /Operation / Time (hour)
- ▷ Actual Working Time (hour)
- ▷ Genealogy Lookup

► Non Conformance:

- ▷ NCR by Work Center
- ▷ NCR by the Reason Code

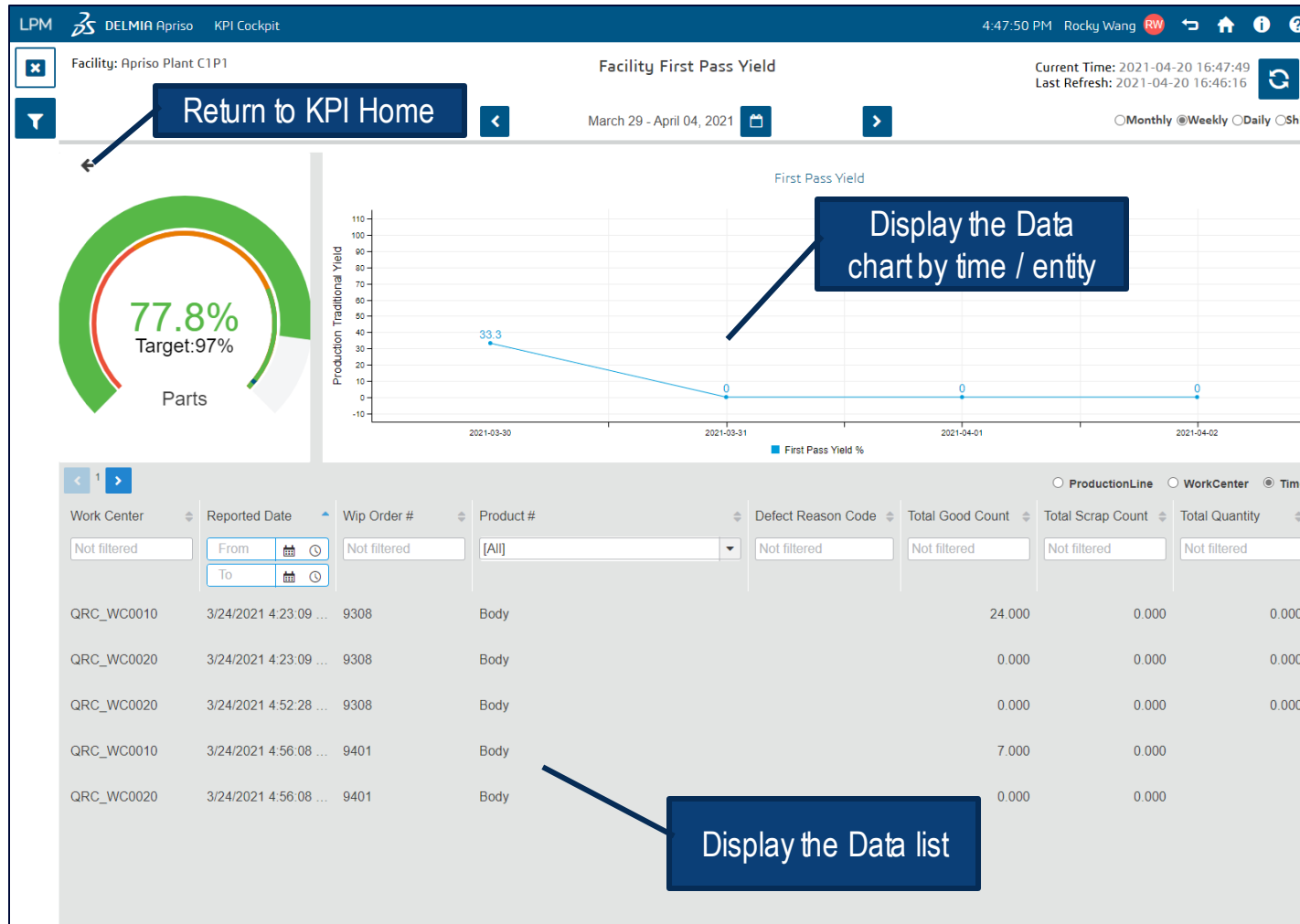
KPI Cockpit

KPI Cockpit can be displayed as the Andon board for shop floor



- Show the KPIs which configured for Facility, PL, Work center, or employee
- Time by Monthly, Weekly, Daily or Shift
- Drill down to show the detail data and analysis

KPI Cockpit - KPI drill down analysis



- ▶ The drill down report is variable by the KPI and selected entity
- ▶ Drill down by time: from month to day, day to hour etc.
- ▶ Drill down by entity, from Facility to production line, work center etc.
- ▶ Show the detail data and chart

Other Dashboard and Reports

- ▶ Machine Availability Analysis
- ▶ Resource Performance Analysis
- ▶ Non Conformance Report
- ▶ Process Data Report
- ▶ Production Report by Weekly
- ▶ Throughout History Report
- ▶ Planning vs. Actual Production Report
- ▶ Production Cycle Time Report
- ▶ Work Center Score Card
- ▶ Downtime Classification Report
- ▶ TEEP Report
- ▶ Line OEE
- ▶ OEE Trend analysis
- ▶ Production Line Performance
- ▶ Andon Board
- ▶ Genealogy Lookup

Automatic test solution - Overview

- The solution is aimed to perform automated tests on Apriso screens as flexparts on browser. The tests are performed based on the inputs and actions provided via **JSON**.
- It is created using C#, .NET Framework 4.7 in Visual Studio 17.
- For creating tests, NUnit Framework is used.



NUnit by Charlie Poole, Rob Prouse



NUnit is a unit-testing framework for all .NET languages with a strong TDD focus.



NUnit.ConsoleRunner by Charlie Poole, Rob Prouse



Console runner for the NUnit 3 unit-testing framework, without any extensions.



NUnit3TestAdapter by Charlie Poole, Terje Sandstrom



NUnit 3 adapter for running tests in Visual Studio. Works with NUnit 3.x, use the NUnit 2 adapter for 2.x tests.

- For automating browser actions, Selenium WebDriver and ChromeDriver is used.



Selenium.Support by Selenium Committers



Provides support classes for Selenium WebDriver



Selenium.WebDriver by Selenium Committers



.NET bindings for the Selenium WebDriver API



Selenium.WebDriver.ChromeDriver by jsakamoto



Selenium Google Chrome Driver (Win32, macOS, and Linux64) (does not make your source repository too fat.)

Help?

- ▶ Line Performance Management Accelerator Help Center
 - ▷ http://ussvcsln20/apriso/help/en-us/svc/svc_lpm_help.htm
- ▶ Release Notes:
 - ▷ Line Performance Management Accelerator A20 Release Notes
(http://ussvcsln20/Apriso/help/en-us/svc/LPM_Release_Notes.htm)